MERN FULL-STACK DEVELOPMENT — OFFICIAL COURSE HANDBOOK

STUDENT GITHUB WORKFLOW & SYLLABUS BRANCHING GUIDE

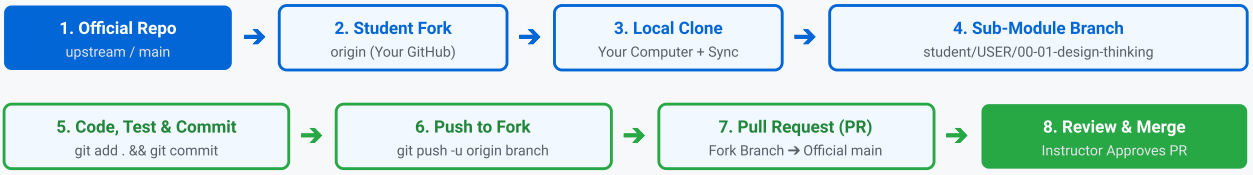
Fork • Sync • Branch • Code • Commit • Push • Pull Request • Review • Merge

Official Repository: https://github.com/fse-brijesh-n/mern-fullstack | Default Branch: `main`

> **Students work in personal forks. The instructor reviews and merges approved pull requests into the official main branch. Never push directly to the instructor's official repository.**

Section A & B — Understand the Complete Step-by-Step Workflow

Every practical exercise in the syllabus follows a strict 9-stage cycle. Understanding how your computer, your GitHub fork, and the instructor's repository interact guarantees smooth submissions without lost code:



Git Term	Beginner-Friendly Meaning in Our Course
`upstream`	The instructor's official repository (`fse-brijesh-n/mern-fullstack`) holding the master syllabus, folders (`BACKEND` , `FRONTEND` , `PROJECTS`), `README.md` , and `mern-handbook.md` .
`origin`	Your personal copy (fork) on your own GitHub account where you have permission to push branches.
`main`	The default clean branch. You sync `main` with the instructor's repo, but you never write assignment code directly on `main` .
Fetch & Pull / Sync	Downloading the latest syllabus updates from `upstream/main` into your local computer and updating your personal GitHub fork.
Branch & Commit	A dedicated workspace for one sub-module (Branch) and saved checkpoints of your code (Commits).
Pull Request (PR)	Your formal submission asking the instructor to review your branch and merge it into the official `main` branch.

Section C — Step-by-Step: Create Your GitHub Account

Follow every step below carefully if you do not yet have a GitHub account:

1. **STEP 1** Open your web browser and go to https://github.com/, then click **Sign up** in the top-right corner.
2. **STEP 2** Enter your active personal or student email address and click **Continue**.
3. **STEP 3** Create a strong password (at least 15 characters, or 8 characters including a number and lowercase letter) and save it securely.
4. **STEP 4** Choose a clean, lowercase **GitHub Username** (for example, `rahul` or `rahul-sharma`). *Remember this exact username—you will use it in every branch name!*
5. **STEP 5** Complete the verification puzzle, open your email inbox, copy the 8-digit launch code sent by GitHub, and paste it into the verification screen.
6. **STEP 6** Once your dashboard opens, click your top-right avatar ➔ **Settings** ➔ **Public profile** and enter your real Full Name so the instructor can identify your submissions.

Section D — Step-by-Step: Install Git Bash & VS Code on Windows

Download and install the two required tools from their official links before writing any commands:

- **Official Git Download:** https://git-scm.com/downloads
- **Official Visual Studio Code Download:** https://code.visualstudio.com/

Windows Git Bash Installation Walkthrough:

1. **STEP 1** Click **Windows** on the Git downloads page and run the downloaded `Git-2.x.x-64-bit.exe` installer.
2. **STEP 2** On the *Select Components* screen, keep **Git Bash Here** checked and click **Next**.
3. **STEP 3** On *Adjusting the name of the initial branch in new repositories*, select **Override the default branch name for new repositories** and make sure it says `main`.
4. **STEP 4** Keep the recommended defaults for PATH environment (*Git from the command line and also from 3rd-party software*), OpenSSH, HTTPS transport, line endings, and make sure **Enable Git Credential Manager** is checked on the final screen, then click **Install**.
5. **STEP 5** Open the Windows Start Menu, type **Git Bash**, and press **Enter** to launch your terminal.
6. **STEP 6** Type the commands below in Git Bash (replacing the quotes with your real name and the exact email linked to your GitHub account) and press **Enter** after each line:

```
 git --version
git config --global user.name "Your Full Name"
git config --global user.email "your-github-email@example.com"
git config --global init.defaultBranch main
git config --global --list
```

Section E — Step-by-Step: Fork the Official Repository

Forking creates your personal copy of the course repository inside your own GitHub account:

1. **STEP 1** Sign in to GitHub in your browser.
2. **STEP 2** Open the official course repository URL: https://github.com/fse-brijesh-n/mern-fullstack.
3. **STEP 3** Look at the top-right action bar of the repository page and click the **Fork** button.
4. **STEP 4** Under **Owner**, select your personal GitHub account, keep the Repository name as `mern-fullstack`, leave *"Copy the main branch only"* checked, and click the green **Create fork** button.
5. **STEP 5** Wait 5 seconds and check your browser address bar. Confirm the URL now contains your username: `https://github.com/GITHUB_USERNAME/mern-fullstack`.

[Diagram Placeholder: GitHub Repository Header Bar]
fse-brijesh-n / mern-fullstack (Public) [Pin] [Watch] [**Fork ▼ (Click Here)**] [Star]
After clicking Create fork, verify header reads: GITHUB_USERNAME / mern-fullstack (forked from fse-brijesh-n/mern-fullstack)

Section F — Step-by-Step: Clone Your Fork & Configure Upstream

Now download your personal fork onto your computer and link it to the instructor's official repository for syllabus updates:

1. **STEP 1** Open **Git Bash** (or the VS Code Terminal) and navigate to the folder where you keep your coding projects (e.g., `cd ~/Desktop`).
2. **STEP 2** Clone your personal fork (replace `GITHUB_USERNAME` with your actual GitHub username):
3. **STEP 3** Enter the project folder using `cd mern-fullstack`.
4. **STEP 4** Add the instructor's official repository as the `upstream` remote and verify with `git remote -v`:

```
git clone https://github.com/GITHUB_USERNAME/mern-fullstack.git
cd mern-fullstack
git remote add upstream https://github.com/fse-brijesh-n/mern-fullstack.git
git remote -v
```

Expected Verification Output from `git remote -v`:
`origin` `https://github.com/GITHUB_USERNAME/mern-fullstack.git (fetch / push)` ➔ Your personal fork.
`upstream` `https://github.com/fse-brijesh-n/mern-fullstack.git (fetch / push)` ➔ Official repo (added only once per clone; if it already exists, inspect it with `git remote -v` instead of adding it again).

Section G — Step-by-Step: How to Sync Your Repository (Fetch, Pull, Merge & Push)

As the course progresses, the instructor updates `README.md`, `mern-handbook.md`, and project templates in the official repository. You must synchronize your local clone and your GitHub fork before starting every new sub-module.

Method 1 — Terminal Sync (Recommended Before Every Sub-Module)

Run these exact commands inside your `mern-fullstack` folder in Git Bash:

```
 # Step 1: Switch to your local main branch
git switch main

# Step 2: Fetch (download) the latest commits and branches from the instructor's official
git fetch upstream

# Step 3: Merge the instructor's latest official main branch into your local main branch
git merge upstream/main

# Step 4: Also pull any updates from your own remote fork's main (if you clicked Sync on G
git pull origin main

# Step 5: Push the newly updated main branch back to your personal GitHub fork (origin)
git push origin main
```

Method 2 — Syncing on GitHub Web ("Sync fork" Button) + Local Pull

1. **STEP 1** Open your fork on GitHub: `https://github.com/GITHUB_USERNAME/mern-fullstack`.
2. **STEP 2** Click the **Sync fork** button located just below the green **Code** button, then click **Update branch**.
3. **STEP 3** Immediately update your computer's local copy by running `git switch main` followed by `git pull origin main` in Git Bash.

Sync Command	What It Does Behind the Scenes
`git fetch upstream`	Contacts `fse-brijesh-n/mern-fullstack` and downloads all new instructor changes safely without touching your working files yet.
`git merge upstream/main`	Applies the downloaded official `upstream/main` updates into your currently active local `main` branch.
`git pull upstream main`	A shortcut that runs `git fetch upstream` + `git merge upstream/main` in a single step.
`git push origin main`	Uploads your freshly synced local `main` branch to your personal GitHub fork so both your computer and your fork match the instructor's repo.

Section H — Mandatory Branch Naming Convention & Creating Your Branch

Every single syllabus sub-module must have its own dedicated branch and its own pull request. Use this exact pattern:

> `student/GITHUB_USERNAME/SUBMODULE-NUMBER-SUBMODULE-NAME`

Strict Branch Naming Rules:

- Replace `GITHUB_USERNAME` with your actual GitHub username (e.g., `rahul`).
- Use the exact sub-module number from the official `mern-handbook.md`, preserving leading zeros (e.g., `00-01`, `01-02`).
- Convert the sub-module title to **lowercase kebab-case**: replace spaces with hyphens (`-`) and remove all punctuation or special characters such as `&`, `:`, or `/`.
- Do not omit, renumber, combine, or invent sub-modules. Create **one branch and one pull request per sub-module**.

Step-by-Step: Create a New Sub-Module Branch

Run this exact sequence in Git Bash every time you begin a new sub-module assignment (shown for Sub-Module `00-01 Design Thinking`):

```
# Step 1: Download latest syllabus updates from official repo
git fetch upstream

# Step 2: Switch to your local main branch
git switch main

# Step 3: Merge official updates into your local main
git merge upstream/main

# Step 4: Create and switch to your dedicated sub-module branch
git switch -c student/GITHUB_USERNAME/00-01-design-thinking

# Step 5: Confirm you are on your new sub-module branch (marked with *)
git branch
```

The command `git switch -c student/GITHUB_USERNAME/00-01-design-thinking` creates the new branch from your synced `main` and switches your workspace onto it immediately.

> **Important Precaution Before Switching Branches:** Always save and commit any unfinished work before running `git switch`. If synchronizing produces merge conflicts, stop immediately—do not run destructive commands—and resolve them carefully or ask your instructor for help.

Section I — Step-by-Step: Complete, Stage, Commit, and Push Your Code

Once your sub-module branch is active, complete your practical exercises inside VS Code, test your solution thoroughly, and push your branch to your personal fork:

1. **STEP 1** Open the project in VS Code (run `code .` inside Git Bash) and complete the assigned exercises inside the appropriate course directory (`FRONTEND` , `BACKEND` , or `PROJECTS`).
2. **STEP 2** Run and test your code locally to verify there are no syntax or runtime errors.
3. **STEP 3** Check which files were modified or created using `git status` .
4. **STEP 4** Stage your files, create a descriptive commit message containing the sub-module number, and push the branch to `origin` (your fork):

```
git status
git add .
git commit -m "Complete 00-01 Design Thinking"
git push -u origin student/GITHUB_USERNAME/00-01-design-thinking
```

Git Command	Step-by-Step Explanation
`git status`	Lists all modified, untracked, and staged files so you can inspect your changes before committing.
`git add .`	Stages all changed files in the current directory for the next commit.
`git commit -m "Complete 00-01..."`	Records a permanent snapshot of your staged work on your local sub-module branch with a clear message including the sub-module number.
`git push -u origin student/...`	Uploads your branch to your personal GitHub fork (`origin`) and links your local branch to the remote branch for future pushes.

Mandatory Security Rule: Always review staged files with `git status` before committing. **Never** commit `.env` files, passwords, database URIs, API keys, access tokens, private credentials, `node_modules/` folders, or unrelated personal files. Note: On your first push, Windows Git Credential Manager will pop up a browser window—click **Authorize GitCredentialManager** to sign in with GitHub.

Section J — Step-by-Step: Submit Your Pull Request (PR) to the Instructor

Pushing code to your fork does not submit it automatically. You must open a Pull Request from your fork's sub-module branch into the official repository's `main` branch:

1. **STEP 1** Open your personal fork in your web browser: `https://github.com/GITHUB_USERNAME/mern-fullstack`.
2. **STEP 2** Click the green **Compare & pull request** banner that appears at the top (or click the **Pull requests** tab ➔ **New pull request**).
3. **STEP 3** Verify the four branch selector dropdowns at the top of the page match this exact configuration table:

Pull Request Field	Required Value
Base repository	`fse-brijesh-n/mern-fullstack`
Base branch	`main`
Head repository	`GITHUB_USERNAME/mern-fullstack`
Compare branch	`student/GITHUB_USERNAME/00-01-design-thinking`
Example title	`00-01 Design Thinking — GITHUB_USERNAME`

Step 4 & Step 5 — Fill Out the Required Pull Request Description & Submit

Copy and fill out the template below inside the PR description box, then click **Create pull request**:

```
### Sub-Module Number & Title
00-01 Design Thinking

### Summary of Completed Work
- Completed user empathy mapping and problem definition exercises for Module 00-01.

### Files or Features Changed
- FRONTEND/00-design-thinking/00-01-design-thinking/README.md

### Commands or Tests Executed
- Verified markdown formatting and asset paths locally in VS Code.

### Known Problems or Questions for Instructor
- None (or list specific questions for review).
```

> **Who Merges the Pull Request?** Only the instructor decides whether and when to merge your Pull Request into the official `main` branch. Never close or merge the PR yourself.

Section K — Step-by-Step: Respond to Instructor Review Feedback

After reviewing your Pull Request, the instructor will either approve and merge it, or leave comments requesting improvements. If changes are requested, **do not create a new branch and do not open a new Pull Request**:

1. **STEP 1** In Git Bash, make sure you are on the exact same sub-module branch: `git switch student/GITHUB_USERNAME/00-01-design-thinking`
2. **STEP 2** Edit your files in VS Code to fix every item mentioned by the instructor and re-test your code.
3. **STEP 3** Stage, commit, and push your fixes to the existing branch:

```
git add .
git commit -m "Address review feedback"
git push
```

Because your branch already tracks `origin`, running `git push` automatically appends your new commit to the **existing open Pull Request** on GitHub so the instructor can re-review it.

Section L — Step-by-Step: Start the Next Sub-Module

Once your previous work has been reviewed/merged and you are ready to begin the next syllabus sub-module (for example, `00-02 UX Research`), synchronize your local `main` and your fork first:

1. **STEP 1** Switch back to your local `main` branch.
2. **STEP 2** Fetch the latest commits from the official `upstream` repository.
3. **STEP 3** Merge `upstream/main` into your local `main` branch.
4. **STEP 4** Push the updated `main` branch to your personal fork (`origin`).
5. **STEP 5** Create a fresh branch for the next sub-module:

```
git switch main
git fetch upstream
git merge upstream/main
git push origin main
git switch -c student/GITHUB_USERNAME/00-02-ux-research
```

> **Strict Sync Safety Rule:** Never use `git push --force`, `git reset --hard`, or overwrite work to fix synchronization issues. If Git reports a conflict during `git merge upstream/main`, ask your instructor for guidance.

Section M — Official Syllabus Branch Naming Reference

Below are all verified sub-modules for **Module 00 — Design Thinking & Figma** (12 sub-modules) and verified examples for **Module 01 — Internet & Web Fundamentals** using `rahul` as the placeholder username:

Module	Sub-Module Number & Title	Exact Branch Name (`rahul`)	Example PR Title
Module 00 — Design Thinking & Figma	00-01 Design Thinking	`student/rahul/00-01-design-thinking`	`00-01 Design Thinking — rahul`
Module 00 — Design Thinking & Figma	00-02 UX Research	`student/rahul/00-02-ux-research`	`00-02 UX Research — rahul`
Module 00 — Design Thinking & Figma	00-03 Information Architecture	`student/rahul/00-03-information-architecture`	`00-03 Information Architecture — rahul`
Module 00 — Design Thinking & Figma	00-04 Wireframing	`student/rahul/00-04-wireframing`	`00-04 Wireframing — rahul`
Module 00 — Design Thinking & Figma	00-05 Figma Fundamentals	`student/rahul/00-05-figma-fundamentals`	`00-05 Figma Fundamentals — rahul`
Module 00 — Design Thinking & Figma	00-06 Figma Advanced	`student/rahul/00-06-figma-advanced`	`00-06 Figma Advanced — rahul`
Module 00 — Design Thinking & Figma	00-07 Design Systems	`student/rahul/00-07-design-systems`	`00-07 Design Systems — rahul`
Module 00 — Design Thinking & Figma	00-08 Prototyping	`student/rahul/00-08-prototyping`	`00-08 Prototyping — rahul`
Module 00 — Design Thinking & Figma	00-09 Handoff	`student/rahul/00-09-handoff`	`00-09 Handoff — rahul`
Module 00 — Design Thinking & Figma	00-10 Accessibility Design	`student/rahul/00-10-accessibility-design`	`00-10 Accessibility Design — rahul`
Module 00 — Design Thinking & Figma	00-11 Visual Design	`student/rahul/00-11-visual-design`	`00-11 Visual Design — rahul`
Module 00 — Design Thinking & Figma	00-12 Design Critique	`student/rahul/00-12-design-critique`	`00-12 Design Critique — rahul`
Module 01 — Internet &		`student/rahul/01-01-how-`	`01-01 How`

Section N — Safe Troubleshooting Guide

Use these safe, non-destructive diagnostic commands if you run into common Git issues:

Problem / Error Message	Safe Diagnostic & Recovery Procedure
1. Git Authentication Failure	GitHub no longer accepts account passwords in the terminal. Run `git push` and sign in via the **Git Credential Manager** browser popup, or generate a Personal Access Token (Classic, `repo` scope) under GitHub Settings ➔ Developer settings.
2. Incorrect `origin` or `upstream`	Run `git remote -v`. Fix incorrect URLs safely using: `git remote set-url origin https://github.com/GITHUB_USERNAME/mern-fullstack.git` `git remote set-url upstream https://github.com/fse-brijesh-n/mern-fullstack.git`
3. "Branch already exists"	If `git switch -c` says the branch already exists, switch to it without `-c`: `git switch student/GITHUB_USERNAME/00-01-design-thinking`
4. "Nothing to commit, working tree clean"	Make sure you pressed `Ctrl + S` inside VS Code to save your files, and verify you are inside the `mern-fullstack` directory using `pwd` and `git status`.
5. Merge Conflicts During Sync	Run `git status` to see conflicted files. If you want to cancel the merge safely and ask your instructor for help, run `git merge --abort`.
6. Wrong Pull Request Base Branch	Open your PR on GitHub, click **Edit** next to the title, and set Base repository to `fse-brijesh-n/mern-fullstack` and Base branch to `main`.
7. Working Tree Is Not Clean	Git prevents switching branches if uncommitted edits would be overwritten. Either commit your changes first, or temporarily shelve them with `git stash` and restore them on your target branch with `git stash pop`.
8. Accidentally Started Work on `main`	Before committing on `main`, immediately move your uncommitted work onto a new sub-module branch by running: `git switch -c student/GITHUB_USERNAME/SUBMODULE-NUMBER-SUBMODULE-NAME`

Section O — Printable Student Submission Checklist

Check off every item below before sharing your Pull Request link with your instructor:

- ☐ [ ] I created my GitHub account and configured my name and email in Git Bash on Windows.
- ☐ [ ] I forked the official repository (`fse-brijesh-n/mern-fullstack`) into my personal GitHub account.
- ☐ [ ] I cloned my personal fork (`origin`) onto my computer.
- ☐ [ ] I configured `upstream` pointing to `https://github.com/fse-brijesh-n/mern-fullstack.git` and verified with `git remote -v`.
- ☐ [ ] I synced my repository using `git fetch upstream`, `git merge upstream/main`, and `git push origin main`.
- ☐ [ ] I created a dedicated branch using the exact official sub-module number and lowercase kebab-case name.
- ☐ [ ] I completed and tested my work and verified `git status` contains no secrets or `.env` files.
- ☐ [ ] I committed and pushed my branch to my personal fork (`origin`).